

COMICCRAFT

AI Comic Story Creator using Gemini Models

PROJECT TESTING & QUALITY VERIFICATION

Input	Generate	Preview	Export	Verify
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Testing is organized around the complete ComicCraft workflow: user input, AI story generation, illustration generation, preview rendering and PDF export.

1. Testing Strategy

The testing approach verifies the application through its main user-facing workflow and checks that routing, input processing, AI service calls, image generation, layout building, preview rendering and PDF export work together.

The supplied ComicCraft project report identifies testing of valid user inputs, route behavior, successful comic generation, image generation, preview rendering and successful PDF export. Invalid inputs and AI/API failures should be handled gracefully.

2. Functional Test Cases

Test Case	Action	Expected Result
TC-01 Home Page	Open the ComicCraft application.	Homepage loads correctly and displays the comic creation form.
TC-02 Valid Input	Enter story prompt, character, setting, tone and art style.	Inputs are accepted and comic generation starts.
TC-03 Outline	Generate the comic outline.	Structured five-panel outline is produced.
TC-04 Story	Expand the outline.	Narration, captions and character dialogue are generated.
TC-05 Image	Generate panel illustrations.	Comic-style images are created and saved.
TC-06 Preview	Open the generated comic preview.	Panel text and corresponding illustrations are displayed.
TC-07 PDF Export	Export the completed comic.	A multi-page PDF is created successfully.

3. Input Validation Tests

Test empty or incomplete story prompts, missing character/setting values, and unsupported or malformed input values. The expected behavior is to reject invalid input or handle it safely without creating an invalid comic.

4. AI & Image Generation Tests

Verify that Gemini Flash produces the structured panel outline, Gemini Pro produces narration and dialogue, and Stable Diffusion produces the requested comic-style illustrations. AI/API failures should be handled gracefully.

5. Preview & Layout Tests

Verify that each generated image is matched with the correct panel story. The preview should display the panel title, illustration, scene description, captions and narration in the expected order.

6. PDF Export Tests

Verify that the layout builder assembles all panels correctly and that the PDF exporter creates a readable multi-page comic containing the generated illustrations and narration.

7. Negative / Failure Tests

Scenario	Expected Result
Invalid user input	Validation error or safe rejection; no invalid comic is created.
AI generation failure	Application handles the failure gracefully.
Image generation failure	Application reports/handles the image-generation error without crashing.
PDF export failure	Application handles export failure safely and does not present a false success.

8. Acceptance Criteria

- Valid inputs successfully generate a comic outline, story, illustrations and preview.
- Generated panel images correspond to the correct panel stories.
- PDF export produces the completed multi-page comic.
- Invalid inputs are handled without creating invalid results.
- AI/API and image-generation failures are handled gracefully.
- The application can be locally tested and verified through its main workflow.

9. Testing Checklist

Check	Status to Verify
Homepage / input form	Loads correctly
Story outline generation	Five-panel structure generated
Story & dialogue	Narration and dialogue generated
Illustrations	Images generated for panels
Comic preview	Text and images aligned
PDF export	Multi-page PDF created
Invalid input handling	Rejected / handled safely
AI/API failure handling	Graceful failure

Reference basis: supplied project testing document, adapted to the ComicCraft project without adding unsupported test results.